Chapter 4. Introduction to Generalized Linear Models

Exponential family, likelihood, deviance, fitting

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Chapters 2–3 treated two-way tables. Real studies have several explanatory variables, some continuous, and we want to describe their effects on a response. This chapter introduces generalized linear models (GLMs), a single framework that contains logistic regression, Poisson loglinear models and ordinary regression. Sections 4.1–4.3 present the models; Sections 4.4–4.6 give the general likelihood theory (likelihood equations, information, deviance, fitting algorithms).

4.1 Generalized linear model

Motivation. Ordinary regression assumes a normal response whose mean is linear in the predictors. For counts and proportions neither assumption is natural; a GLM keeps the linear predictor but lets the response distribution and the scale on which linearity holds be chosen separately.

4.1.1 Components of generalized linear models

Definition (Natural exponential family)

A family of densities or mass functions of the form

$$f(y_i;\theta_i) = a(\theta_i)\, b(y_i) \exp\{y_i\, Q(\theta_i)\} \tag{4.1}$$

is a *natural exponential family*; $Q(\theta)$ is the *natural parameter*. The parameter θ_i may vary with i through explanatory variables.

Definition (Generalized linear model)

A GLM has three components.
(i) *Random component*: independent observations $y_1, \ldots, y_N$ of Y, each from a natural exponential family (4.1) (more generally, the exponential dispersion family of Section 4.4).
(ii) *Systematic component*: a *linear predictor* $\eta_i = \sum_{j=1}^{p} \beta_j x_{ij}$, $i = 1, \ldots, N$, where x_{ij} is the value of predictor j for subject i (usually $x_{i1} \equiv 1$ gives an intercept, often written α).
(iii) *Link function*: a monotone differentiable g with

$$g(\mu_i) = \eta_i = \sum_j \beta_j x_{ij}, \qquad \mu_i = E(Y_i). \tag{4.2}$$

The *identity link* $g(\mu) = \mu$ gives a linear model for the mean. The link with $g(\mu_i) = Q(\theta_i)$, transforming the mean to the natural parameter, is the *canonical link*.

In short, a GLM is a linear model for a transformed mean of a response whose distribution lies in the natural exponential family.

4.1.2 Binomial logit models for binary data

Code a binary outcome as 1 (success) and 0 (failure), with $P(Y = 1) = \pi = E(Y)$. The Bernoulli mass function is

$$f(y;\pi) = \pi^y (1-\pi)^{1-y} = (1-\pi) \exp\Big[y \log \frac{\pi}{1-\pi}\Big], \qquad y = 0, 1,$$

which is (4.1) with $\theta = \pi$, $a(\pi) = 1-\pi$, $b(y) = 1$ and $Q(\pi) = \log[\pi/(1-\pi)]$. The natural parameter is the log odds, the *logit* of π; the logit is the canonical link, and GLMs with this link are *logit models*.

4.1.3 Poisson loglinear models for count data

For a count Y with mean μ,

$$f(y;\mu) = \frac{e^{-\mu}\mu^{y}}{y!} = \exp(-\mu)\Big(\frac{1}{y!}\Big)\exp(y\log\mu), \qquad y = 0, 1, 2, \ldots,$$

so $\theta = \mu$, $a(\mu) = e^{-\mu}$, $b(y) = 1/y!$, $Q(\mu) = \log\mu$. The canonical link is the log link, and

$$\log\mu_i = \sum_j \beta_j x_{ij}, \qquad i = 1, \ldots, N, \tag{4.3}$$

is the *Poisson loglinear model*.

4.1.4 Generalized linear models for continuous responses

The normal distribution belongs to an exponential family with a dispersion parameter, and its natural parameter is the mean; so ordinary regression, ANOVA and ANCOVA are GLMs with normal random component and identity link.

Random component	Link	Systematic component	Model
Normal	identity	continuous	regression
Normal	identity	categorical	analysis of variance
Normal	identity	mixed	analysis of covariance
Binomial	logit	mixed	logistic regression (Ch. 5–6)
Poisson	log	mixed	loglinear (Ch. 8–9)
Multinomial	generalized logit	mixed	multinomial response (Ch. 7)

The traditional alternative transforms Y to approximate normality with constant variance and applies least squares. In a GLM the choice of link is separate from the choice of random component: a link need only make the linear predictor plausible; it need not stabilize variance or produce normality, because fitting maximizes the likelihood of the chosen distribution.

4.1.5 Deviance

Let $L(\boldsymbol{\mu};\mathbf{y})$ be the log likelihood expressed through the means $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_N)$ and $L(\hat{\boldsymbol{\mu}};\mathbf{y})$ its maximum under a model. Over all models the largest achievable value is $L(\mathbf{y};\mathbf{y})$, attained by the *saturated model*, which has one parameter per observation and fits perfectly ($\hat{\mu}_i = y_i$). The saturated model gives no data reduction but is a baseline for comparison.

Definition (Deviance)

For a Poisson or binomial GLM the *deviance* is

$$D(\mathbf{y};\hat{\boldsymbol{\mu}}) = -2\big[L(\hat{\boldsymbol{\mu}};\mathbf{y}) - L(\mathbf{y};\mathbf{y})\big],$$

the likelihood-ratio statistic for testing that the model holds against the saturated alternative.

When N stays fixed while the individual counts grow, the deviance is asymptotically χ^2 with $\mathrm{df} = N - p$, the difference in the numbers of parameters of the saturated and the fitted model (Theorem 4.19).

Proposition 4.1 (Deviance of the homogeneity model)

Let $y_i \sim \text{bin}(n_i, \pi_i)$ independently, $i = 1, \ldots, N$, and consider the model $\pi_i = \pi$ for all i ($p = 1$). Its deviance equals the statistic G^2 for testing independence in the $N \times 2$ table of successes and failures, and has df $= N - 1$.

Proof.

Step 1. Write $n = \sum n_i$ and $s = \sum y_i$ (here y_i is a count). Up to a constant, $L = \sum_i [y_i \log \pi_i + (n_i - y_i) \log(1 - \pi_i)]$. Under homogeneity the MLE is $\hat{\pi} = s/n$; in the saturated model $\tilde{\pi}_i = y_i/n_i$.

Step 2. Hence

$$D = 2 \sum_i \Big[y_i \log \frac{y_i/n_i}{\hat{\pi}} + (n_i - y_i) \log \frac{1 - y_i/n_i}{1 - \hat{\pi}} \Big]$$
$$= 2 \sum_i \Big[y_i \log \frac{y_i}{n_i s/n} + (n_i - y_i) \log \frac{n_i - y_i}{n_i(n - s)/n} \Big].$$

Step 3. In the $N \times 2$ table with row totals n_i and column totals s, $n - s$, the fitted values under independence are $n_i s/n$ and $n_i(n - s)/n$. So $D = 2 \sum \text{observed} \log(\text{observed}/\text{fitted}) = G^2$. The saturated model has N parameters and the model one, giving df $= N - 1$. ■

Components of the deviance serve as residuals, and comparing deviances generalizes the analysis of variance (Section 4.5).

4.1.6 Advantages of the GLM formulation

GLMs give a unified theory for the most important continuous and discrete models. A single algorithm, iteratively reweighted least squares (Section 4.6), computes the ML estimates for every member of the exponential family and every link; this is the reason for restricting the random component to that family.

4.2 Generalized linear models for binary data

Motivation. For a binary response the mean is a probability $\pi(\mathbf{x}) = P(Y = 1)$ and $\text{Var}(Y) = \pi(\mathbf{x})[1 - \pi(\mathbf{x})]$. Which function of $\pi(\mathbf{x})$ should be linear in $\mathbf{x}$? The answer (identity, logit, probit) determines how effects are interpreted.

We use a single predictor x for simplicity.

4.2.1 Linear probability model

The *linear probability model* $\pi(x) = \alpha + \beta x$ is the binomial GLM with identity link. Its structural defect: probabilities lie in $[0, 1]$ while $\alpha + \beta x$ eventually leaves that range; fitting can fail when $\hat{\pi}(x)$ falls outside $[0, 1]$ for some observed x. Over a restricted range it can be adequate, and then β has the simple meaning "change in probability per unit of x".

It should be fitted by ML with the binomial distribution, not by least squares: since $\text{Var}(Y) = \pi(x)[1 - \pi(x)]$ depends on x, the constant-variance condition making least squares optimal fails, the binomial ML estimator is more efficient, and Y is far from normal. ML and least squares agree closely only when the fitted probabilities stay in about 0.3–0.7, where the variance is nearly constant.

4.2.2 Snoring and heart disease example

Example 4.2 (Snoring and heart disease)

An epidemiological survey of 2484 subjects classified snoring level (reported by the spouse) and heart disease. With scores $x = (0, 2, 4, 5)$ for the snoring levels (the last two treated as closer), the rows are independent binomial samples.

	Heart disease		Proportion	Linear	Logit
Snoring	Yes	No	yes	fit	fit
Never ($x = 0$)	24	1355	0.017	0.017	0.021
Occasionally ($x = 2$)	35	603	0.055	0.057	0.044
Nearly every night ($x = 4$)	21	192	0.099	0.096	0.093
Every night ($x = 5$)	30	224	0.118	0.116	0.132

The ML fit of the linear probability model is $\hat{\pi}(x) = 0.0172 + 0.0198x$ ($\widehat{\text{se}}(\hat{\beta}) = 0.0028$). The estimated risk is about 0.02 for nonsnorers and rises by $2(0.0198) \approx 0.04$ for occasional snorers, another 0.04 for those snoring nearly every night and another 0.02 for those snoring every night. The ML estimates are the same whether the data are entered as 2484 binary observations or as the four binomial totals. The logistic fit (Section 4.2.3) is $\text{logit}\,\hat{\pi}(x) = -3.87 + 0.40x$; over this narrow range of probabilities the two fits are similar (Figure 1).

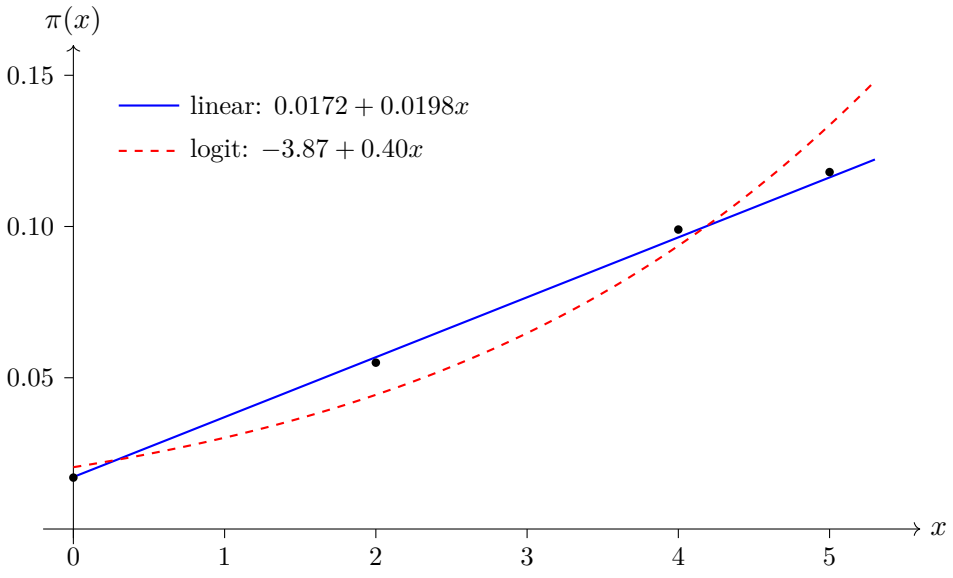


Figure 1: Snoring data: sample proportions (dots) with the linear probability and logistic regression fits.

4.2.3 Logistic regression model

Usually $\pi(x)$ is nonlinear in x: a fixed change in x matters less when $\pi(x)$ is near 0 or 1 than near 0.5. Monotone S-shaped curves are typical, the most important being

$$\pi(x) = \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)}. \tag{4.4}$$

As $x \to \infty$, $\pi(x) \downarrow 0$ if $\beta < 0$ and $\pi(x) \uparrow 1$ if $\beta > 0$.

Proposition 4.3 (Logistic regression is the binomial GLM with logit link)

Model (4.4) holds if and only if

$$\log \frac{\pi(x)}{1-\pi(x)} = \alpha + \beta x.$$

Hence logistic regression is the binomial GLM with the canonical (logit) link.

Proof. From (4.4), $1-\pi(x) = 1/[1+\exp(\alpha+\beta x)]$, so the odds are $\pi(x)/[1-\pi(x)] = \exp(\alpha+\beta x)$; take logs. Conversely, solving $\log[\pi/(1-\pi)] = \eta$ for π gives $\pi = e^{\eta}/(1+e^{\eta})$. The logit is the binomial natural parameter (Section 4.1.2). ■

The logit ranges over $\mathbb{R}$, matching the range of $\alpha+\beta x$, so logistic regression does not have the structural problem of the linear probability model. For the snoring data $\hat{\beta} = 0.40 > 0$ reflects increasing risk with snoring.

4.2.4 Binomial GLM for 2×2 contingency tables

Proposition 4.4 (Effect parameters for a binary predictor)

Let $x \in \{0,1\}$ and $\text{link}[\pi(x)] = \alpha + \beta x$. Then $\beta = \text{link}[\pi(1)] - \text{link}[\pi(0)]$; in particular
(a) identity link: $\beta = \pi(1) - \pi(0)$, the difference of proportions;
(b) log link: $\beta = \log[\pi(1)/\pi(0)]$, the log relative risk;
(c) logit link: $\beta = \log \dfrac{\pi(1)/[1-\pi(1)]}{\pi(0)/[1-\pi(0)]}$, the log odds ratio.

Proof. Subtract $\text{link}[\pi(0)] = \alpha$ from $\text{link}[\pi(1)] = \alpha + \beta$ and specialize the link; for (c), $\text{logit}\,\pi(1) - \text{logit}\,\pi(0)$ is a difference of log odds, i.e. a log odds ratio. ■

Thus the association measures of Chapter 2 are effect parameters of binary-response GLMs.

4.2.5 Probit and inverse cdf link functions

A monotone increasing curve has the shape of a cdf, suggesting $\pi(x) = F(x)$ for a continuous cdf F. Taking a standard cdf Φ of a location–scale class and writing

$$\pi(x) = \Phi(\alpha+\beta x) \quad \Longleftrightarrow \quad \Phi^{-1}[\pi(x)] = \alpha + \beta x \tag{4.5}$$

gives all members of the class: β sets the rate of increase (or decrease when $\beta < 0$) and α shifts the curve. The link is Φ^{-1}, mapping $(0,1)$ onto $(-\infty,\infty)$. With Φ the standard normal cdf, (4.5) is the *probit model* (Section 6.6).

The logistic distribution with location μ and scale $\tau > 0$ has cdf

$$F(x) = \frac{\exp[(x-\mu)/\tau]}{1+\exp[(x-\mu)/\tau]}, \qquad -\infty < x < \infty;$$

its density is symmetric and bell-shaped with standard deviation $\tau\pi/\sqrt{3}$, slightly thicker-tailed than the normal (kurtosis like a t_9). For the standard logistic cdf $\Phi(x) = e^x/(1+e^x)$ we have $\Phi^{-1}(\pi) = \log[\pi/(1-\pi)]$, so the logit link is the inverse cdf link of the logistic distribution, and (4.4) with $\beta > 0$ is a logistic cdf.

4.3 Generalized linear models for counts

Motivation. Counts are nonnegative and their variance typically grows with the mean. The Poisson GLM with log link is the basic model; overdispersion leads to the negative binomial and, later, to quasi-likelihood.

4.3.1 Poisson loglinear models

The log mean is the Poisson natural parameter, so the log link is canonical. With one predictor,

$$\log \mu = \alpha + \beta x, \qquad \mu = \exp(\alpha + \beta x) = e^{\alpha}(e^{\beta})^{x}. \tag{4.6}$$

A one-unit increase in x multiplies the mean by e^{β}.

4.3.2 Horseshoe crab mating example

Example 4.5 (Horseshoe crab satellites)

For 173 nesting female horseshoe crabs the response is her number of *satellites* (additional males residing nearby); predictors are color (1 light medium, 2 medium, 3 dark medium, 4 dark), spine condition, weight (kg) and carapace width (cm). Mean width is 26.3 cm (sd 2.1). Grouping width into eight categories gives:

Width (cm)	Cases	Satellites	Sample mean	Sample variance
< 23.25	14	14	1.00	2.77
23.25–24.25	14	20	1.43	8.88
24.25–25.25	28	67	2.39	6.54
25.25–26.25	39	105	2.69	11.38
26.25–27.25	22	63	2.86	6.88
27.25–28.25	24	93	3.87	8.81
28.25–29.25	18	71	3.94	16.88
> 29.25	14	72	5.14	8.29

The means increase roughly linearly with width. Using the ungrouped data and $x =$ width:
(a) Log link: $\log \hat{\mu} = -3.305 + 0.164x$ ($\widehat{\text{se}} = 0.020$). At $x = 26.3$, $\hat{\mu} = \exp(-3.305 + 0.164 \cdot 26.3) = 2.74$; $e^{0.164} = 1.18$, so a 1-cm increase raises the estimated mean by 18% (at $x = 27.3$, $\hat{\mu} = 3.23 = 1.18 \times 2.74$).
(b) Identity link: $\hat{\mu} = -11.53 + 0.55x$ ($\widehat{\text{se}} = 0.06$). On average about 2 cm more width brings one more satellite.
The two fits diverge at extreme widths but agree where most data lie.

4.3.3 Overdispersion for Poisson GLMs

In the table above the sample variances far exceed the means, whereas a Poisson has equal mean and variance. Greater variability than the random component predicts is *overdispersion*. A common cause is unobserved heterogeneity.

Proposition 4.6 (Mixtures of Poissons are overdispersed)

If $Y \mid \lambda \sim \text{Poisson}(\lambda)$ and λ is random with $E(\lambda) = \mu$, then $E(Y) = \mu$ and $\text{Var}(Y) = \mu + \text{Var}(\lambda) \geq \mu$, with equality only when λ is degenerate.

Proof. [Supplement] By iterated expectations $E(Y) = E[E(Y \mid \lambda)] = E(\lambda) = \mu$, and by the law of total variance

$$\mathrm{Var}(Y) = E[\mathrm{Var}(Y \mid \lambda)] + \mathrm{Var}[E(Y \mid \lambda)] = E(\lambda) + \mathrm{Var}(\lambda) = \mu + \mathrm{Var}(\lambda).$$

■

If, say, Y is Poisson given width, weight, color and spine, then crabs of a given width form a mixture over the other variables, and Y given width alone is overdispersed. When the mean model is right but the distribution is not Poisson, the ML estimates remain consistent but the standard errors are wrong. Overdispersion is not an issue for normal regression, which has a separate variance parameter.

4.3.4 Negative binomial GLMs

The negative binomial mass function is

$$f(y; k, \mu) = \frac{\Gamma(y+k)}{\Gamma(k)\Gamma(y+1)}\Big(\frac{k}{\mu+k}\Big)^k\Big(1-\frac{k}{\mu+k}\Big)^y, \qquad y = 0, 1, 2, \ldots \tag{4.7}$$

Proposition 4.7 (Negative binomial as a gamma mixture)

If $Y \mid \lambda \sim \mathrm{Poisson}(\lambda)$ and λ has the gamma density with shape k and mean μ, then Y has the mass function (4.7), with

$$E(Y) = \mu, \qquad \mathrm{Var}(Y) = \mu + \mu^2/k.$$

For fixed k, (4.7) is a natural exponential family with natural parameter $\log[\mu/(\mu+k)]$, so negative binomial regression is a GLM. As $k^{-1} \to 0$ it converges to the Poisson.

Proof. [Supplement]

Step 1. The gamma density is $h(\lambda) = (k/\mu)^k \lambda^{k-1} e^{-k\lambda/\mu}/\Gamma(k)$, $\lambda > 0$. Then

$$\begin{aligned} P(Y=y) &= \int_0^\infty \frac{e^{-\lambda}\lambda^y}{y!}\, h(\lambda)\, d\lambda = \frac{(k/\mu)^k}{\Gamma(k)\, y!}\int_0^\infty \lambda^{y+k-1} e^{-\lambda(1+k/\mu)} d\lambda \\ &= \frac{\Gamma(y+k)}{\Gamma(k)\, y!}\Big(\frac{k}{\mu}\Big)^k\Big(\frac{\mu}{\mu+k}\Big)^{y+k} = \frac{\Gamma(y+k)}{\Gamma(k)\, y!}\Big(\frac{k}{\mu+k}\Big)^k\Big(\frac{\mu}{\mu+k}\Big)^y, \end{aligned}$$

which is (4.7) since $1 - k/(\mu+k) = \mu/(\mu+k)$.

Step 2. The gamma law has mean μ and variance μ^2/k, so Proposition 4.6 gives $E(Y) = \mu$ and $\mathrm{Var}(Y) = \mu + \mu^2/k$.

Step 3. With $Q = \log[\mu/(\mu+k)]$, (4.7) reads $a\, b(y) \exp(yQ)$ with $a = (1-e^Q)^k$ and $b(y) = \Gamma(y+k)/[\Gamma(k)y!]$, the form (4.1).

Step 4. As $k \to \infty$, $\Gamma(y+k)/[\Gamma(k)(\mu+k)^y] \to 1$ and $(1+\mu/k)^{-k} \to e^{-\mu}$, so $f(y) \to e^{-\mu}\mu^y/y!$. ■

The index k^{-1} is a *dispersion parameter*, usually unknown and common to all observations; log and identity links are both used (Section 13.4).

Example 4.8 (Crab satellites: negative binomial fit)

With identity link and width predictor, the Poisson GLM gives $\hat{\mu} = -11.53 + 0.55x$ ($\widehat{\mathrm{se}} = 0.06$) and the negative binomial GLM gives $\hat{\mu} = -11.15 + 0.53x$ ($\widehat{\mathrm{se}} = 0.11$) with $\hat{k}^{-1} = 0.98$, so the estimated variance is about $\hat{\mu} + \hat{\mu}^2$ rather than $\hat{\mu}$. Fitted values are similar, but the larger

standard error reflects the overdispersion the Poisson model ignores.

4.3.5 Poisson regression for rates

When events occur over time, space or population of varying size t_i, the rate μ_i/t_i is more relevant than the count (e.g. homicides per city population). The loglinear model $\log(\mu_i/t_i) = \sum_j \beta_j x_{ij}$ is $\log \mu_i = \log t_i + \sum_j \beta_j x_{ij}$, a Poisson GLM with the known *offset* $\log t_i$ (Section 9.7).

4.3.6 Poisson GLM of independence in $I \times J$ contingency tables

Proposition 4.9 (Poisson form of independence)

Let Y_{ij} be independent Poisson with means $\mu_{ij} = \mu\alpha_i\beta_j$, where $\alpha_i, \beta_j > 0$ and $\sum_i \alpha_i = \sum_j \beta_j = 1$; equivalently

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y, \qquad \lambda = \log \mu,\ \lambda_i^X = \log \alpha_i,\ \lambda_j^Y = \log \beta_j. \tag{4.8}$$

Conditional on $n = \sum_{i,j} Y_{ij}$, the cell counts are multinomial with probabilities $\pi_{ij} = \alpha_i\beta_j$, which satisfy $\pi_{i+} = \alpha_i$, $\pi_{+j} = \beta_j$ and $\pi_{ij} = \pi_{i+}\pi_{+j}$.

Proof.

Step 1. A sum of independent Poissons is Poisson, so $n \sim \text{Poisson}(\sum \mu_{ij}) = \text{Poisson}(\mu)$.

Step 2. For counts $\{n_{ij}\}$ with total n,

$$P(\{Y_{ij} = n_{ij}\} \mid n) = \frac{\prod_{ij} e^{-\mu_{ij}} \mu_{ij}^{n_{ij}}/n_{ij}!}{e^{-\mu}\mu^n/n!} = \frac{n!}{\prod_{ij} n_{ij}!} \prod_{ij} \Big(\frac{\mu_{ij}}{\mu}\Big)^{n_{ij}},$$

a multinomial with $\pi_{ij} = \mu_{ij}/\mu = \alpha_i\beta_j$.

Step 3. Summing, $\pi_{i+} = \alpha_i \sum_j \beta_j = \alpha_i$ and $\pi_{+j} = \beta_j$, so $\pi_{ij} = \pi_{i+}\pi_{+j}$. ■

So the Chapter 3 inferences about independence are inferences for the loglinear model (4.8) or the corresponding multinomial model; Chapters 8–9 develop richer loglinear models.

4.4 Moments and likelihood for generalized linear models

Motivation. All GLM computations rest on a few facts about the exponential dispersion family: the mean and variance come from derivatives of one function $b(\theta)$, and the likelihood equations and the information matrix depend on the distribution only through the mean and variance.

Definition (Exponential dispersion family)

The observations $y_1, \ldots, y_N$ are independent with density or mass function

$$f(y_i; \theta_i, \phi) = \exp\Big\{\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\Big\}. \tag{4.9}$$

θ_i is the *natural parameter* and ϕ the *dispersion parameter*. Usually $a(\phi) = \phi/\omega_i$ with a known weight ω_i (e.g. $\omega_i = n_i$ when y_i is a mean of n_i readings).

For known ϕ, (4.9) is the natural exponential family (4.1) with $Q(\theta) = \theta/a(\phi)$, $a(\theta) = \exp[-b(\theta)/a(\phi)]$ and $b(y) = \exp[c(y, \phi)]$.

4.4.1 Mean and variance functions for the random component

Let $L_i = \log f(y_i; \theta_i, \phi)$, so the log likelihood is $L = \sum_i L_i$ and

$$L_i = \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi). \tag{4.10}$$

Theorem 4.10 (Moments of the exponential dispersion family)

For (4.9), with θ_i interior to the natural parameter space,

$$\mu_i = E(Y_i) = b'(\theta_i), \qquad \mathrm{Var}(Y_i) = b''(\theta_i)\, a(\phi). \tag{4.11}$$

Proof. [Supplement]

Step 1. (Bartlett identities.) Write $f = f(y;\theta)$ and $\ell = \log f$. Because $\int f\, dy = 1$ for all θ and, for an exponential family with θ in the interior of the parameter space, differentiation under the integral sign is permitted (the support does not depend on θ and the integrand has locally dominated derivatives),

$$0 = \frac{d}{d\theta}\int f\, dy = \int \frac{\partial \ell}{\partial \theta} f\, dy = E\Big(\frac{\partial \ell}{\partial \theta}\Big).$$

Differentiating once more, using $\partial f/\partial\theta = f\,\partial\ell/\partial\theta$,

$$0 = \int \Big[\frac{\partial^2 \ell}{\partial \theta^2} + \Big(\frac{\partial \ell}{\partial \theta}\Big)^2\Big] f\, dy, \quad \text{i.e.} \quad -E\Big(\frac{\partial^2 \ell}{\partial \theta^2}\Big) = E\Big(\frac{\partial \ell}{\partial \theta}\Big)^2.$$

(For discrete Y replace integrals by sums.)

Step 2. From (4.10),

$$\frac{\partial L_i}{\partial \theta_i} = \frac{y_i - b'(\theta_i)}{a(\phi)}, \qquad \frac{\partial^2 L_i}{\partial \theta_i^2} = -\frac{b''(\theta_i)}{a(\phi)}.$$

Step 3. The first identity gives $E[Y_i - b'(\theta_i)]/a(\phi) = 0$, i.e. $\mu_i = b'(\theta_i)$.

Step 4. The second identity gives

$$\frac{b''(\theta_i)}{a(\phi)} = E\Big[\frac{(Y_i - b'(\theta_i))^2}{a(\phi)^2}\Big] = \frac{\mathrm{Var}(Y_i)}{a(\phi)^2},$$

hence $\mathrm{Var}(Y_i) = b''(\theta_i)a(\phi)$. ∎

Since $\mathrm{Var}(Y_i) > 0$, $b'' > 0$: b' is strictly increasing, so θ_i and μ_i determine each other. The function b is the *cumulant function*.

4.4.2 Mean and variance functions for Poisson and binomial

Proposition 4.11 (Poisson and binomial in dispersion form)

(a) Poisson: $\theta_i = \log \mu_i$, $b(\theta_i) = e^{\theta_i}$, $a(\phi) = 1$, $c = -\log y_i!$; then $E(Y_i) = \mathrm{Var}(Y_i) = \mu_i$.
(b) Binomial proportion: if $n_i Y_i \sim \mathrm{bin}(n_i, \pi_i)$, then $\theta_i = \log[\pi_i/(1-\pi_i)]$, $b(\theta_i) = \log(1 + e^{\theta_i})$, $a(\phi) = 1/n_i$, $c = \log \binom{n_i}{n_i y_i}$; then $E(Y_i) = \pi_i$ and $\mathrm{Var}(Y_i) = \pi_i(1-\pi_i)/n_i$.

Proof.

(a) $f(y_i;\mu_i) = e^{-\mu_i}\mu_i^{y_i}/y_i! = \exp[y_i\theta_i - e^{\theta_i} - \log y_i!]$. By Theorem 4.10, $E(Y_i) = b'(\theta_i) = e^{\theta_i} = \mu_i$ and $\mathrm{Var}(Y_i) = b''(\theta_i) = \mu_i$.

(b) With $\pi_i = e^{\theta_i}/(1+e^{\theta_i})$ we have $\log(1-\pi_i) = -\log(1+e^{\theta_i})$, so

$$f(y_i;\pi_i,n_i) = \binom{n_i}{n_i y_i}\pi_i^{n_i y_i}(1-\pi_i)^{n_i - n_i y_i} = \binom{n_i}{n_i y_i}\exp\{n_i[y_i\theta_i + \log(1-\pi_i)]\}$$
$$= \exp\Big[\frac{y_i\theta_i - \log(1+e^{\theta_i})}{1/n_i} + \log\binom{n_i}{n_i y_i}\Big].$$

Then $b'(\theta_i) = e^{\theta_i}/(1+e^{\theta_i}) = \pi_i$ and $b''(\theta_i)a(\phi) = \dfrac{e^{\theta_i}}{(1+e^{\theta_i})^2}\cdot\dfrac{1}{n_i} = \dfrac{\pi_i(1-\pi_i)}{n_i}$. ■

Normal case

For $N(\mu_i,\sigma^2)$: $\theta_i = \mu_i$, $b(\theta) = \theta^2/2$, $a(\phi) = \phi = \sigma^2$, giving $E(Y_i) = \mu_i$ and $\mathrm{Var}(Y_i) = \sigma^2$.

4.4.3 Systematic component and link function

In matrix form the linear predictor is $\boldsymbol{\eta} = X\boldsymbol{\beta}$, with $\boldsymbol{\eta} = (\eta_1,\ldots,\eta_N)^\top$, $\boldsymbol{\beta} = (\beta_1,\ldots,\beta_p)^\top$ and X the $N \times p$ *model matrix*; the link gives $\eta_i = g(\mu_i)$.

Proposition 4.12 (Canonical link)

The canonical link (the g with $g(\mu_i) = \theta_i$) is $g = (b')^{-1}$. For the Poisson it is the log link; for the binomial it is the logit link.

Proof. By Theorem 4.10, $\mu_i = b'(\theta_i)$ with b' strictly increasing, so $\theta_i = (b')^{-1}(\mu_i)$. Poisson: $b'(\theta) = e^\theta$, inverse log. Binomial: $b'(\theta) = e^\theta/(1+e^\theta)$, inverse logit. ■

With the canonical link, $\theta_i = \sum_j \beta_j x_{ij}$: the natural parameter itself is linear in the predictors.

4.4.4 Likelihood equations for a GLM

The log likelihood is

$$L(\boldsymbol{\beta}) = \sum_i L_i = \sum_i \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + \sum_i c(y_i,\phi), \tag{4.12}$$

which depends on $\boldsymbol{\beta}$ through $\theta_i = \theta_i(\mu_i)$, $\mu_i = g^{-1}(\eta_i)$, $\eta_i = \sum_j \beta_j x_{ij}$.

Theorem 4.13 (GLM likelihood equations)

For a GLM with exponential dispersion random component,

$$\frac{\partial L_i}{\partial \beta_j} = \frac{(y_i-\mu_i)x_{ij}}{\mathrm{Var}(Y_i)}\frac{\partial \mu_i}{\partial \eta_i}, \tag{4.13}$$

and the likelihood equations are

$$\sum_{i=1}^N \frac{(y_i-\mu_i)x_{ij}}{\mathrm{Var}(Y_i)}\frac{\partial \mu_i}{\partial \eta_i} = 0, \qquad j = 1,\ldots,p. \tag{4.14}$$

Proof.

Step 1. By the chain rule,

$$\frac{\partial L_i}{\partial \beta_j} = \frac{\partial L_i}{\partial \theta_i}\frac{\partial \theta_i}{\partial \mu_i}\frac{\partial \mu_i}{\partial \eta_i}\frac{\partial \eta_i}{\partial \beta_j}.$$

Step 2. From the proof of Theorem 4.10, $\partial L_i/\partial\theta_i = [y_i - b'(\theta_i)]/a(\phi) = (y_i-\mu_i)/a(\phi)$.

Step 3. Since $\mu_i = b'(\theta_i)$, $\partial\mu_i/\partial\theta_i = b''(\theta_i) = \mathrm{Var}(Y_i)/a(\phi)$, so $\partial\theta_i/\partial\mu_i = a(\phi)/\mathrm{Var}(Y_i)$.

Step 4. $\partial\eta_i/\partial\beta_j = x_{ij}$, and $\partial\mu_i/\partial\eta_i = 1/g'(\mu_i)$ depends on the link.

Step 5. Multiplying, $a(\phi)$ cancels and (4.13) follows; summing over i and setting to zero gives (4.14). ∎

Although $\boldsymbol{\beta}$ does not appear explicitly, it enters through $\mu_i = g^{-1}(\sum_j \beta_j x_{ij})$. Crucially, (4.14) depends on the distribution only through μ_i and the *variance function* $\mathrm{Var}(Y_i) = v(\mu_i)$, e.g. $v(\mu)=\mu$ (Poisson), $v(\mu)=\mu(1-\mu)$ (Bernoulli), $v(\mu)=\sigma^2$ (normal). Within the natural exponential family, the mean–variance relation characterizes the distribution (Jorgensen 1987): $v(\mu)=\mu$ forces the Poisson.

4.4.5 Likelihood equations for binomial GLMs

Corollary 4.14 (Binomial likelihood equations)

Let $n_iY_i \sim \mathrm{bin}(n_i,\pi_i)$ and $\pi_i = \Phi(\eta_i)$, $\eta_i = \sum_j \beta_j x_{ij}$, for a continuous cdf Φ with density ϕ. The likelihood equations are

$$\sum_i \frac{n_i(y_i-\pi_i)\,x_{ij}}{\pi_i(1-\pi_i)}\,\phi(\eta_i) = 0, \qquad j = 1,\dots,p. \tag{4.15}$$

For the logit link they reduce to

$$\sum_i n_i(y_i-\pi_i)x_{ij} = 0, \qquad j = 1,\dots,p. \tag{4.16}$$

Proof. Here $\partial\pi_i/\partial\eta_i = \phi(\eta_i)$ and $\mathrm{Var}(Y_i) = \pi_i(1-\pi_i)/n_i$; substitute in (4.14). For the logit, $\eta_i = \log[\pi_i/(1-\pi_i)]$ gives $\partial\eta_i/\partial\pi_i = 1/[\pi_i(1-\pi_i)]$, so $\phi(\eta_i) = \pi_i(1-\pi_i)$ cancels the denominator. ∎

4.4.6 Asymptotic covariance matrix of model parameter estimators

Theorem 4.15 (Information matrix of a GLM)

The expected information matrix $\mathcal{I}$ with entries $E[-\partial^2 L(\boldsymbol{\beta})/\partial\beta_h\partial\beta_j]$ is

$$\mathcal{I} = X^\top W X, \qquad W = \mathrm{diag}(w_1,\dots,w_N), \quad w_i = \frac{(\partial\mu_i/\partial\eta_i)^2}{\mathrm{Var}(Y_i)}. \tag{4.17}$$

Proof.

Step 1. By the second Bartlett identity (proof of Theorem 4.10, applied to $\boldsymbol{\beta}$; valid under the same regularity),

$$E\Big(\frac{\partial^2 L_i}{\partial\beta_h\partial\beta_j}\Big) = -E\Big(\frac{\partial L_i}{\partial\beta_h}\,\frac{\partial L_i}{\partial\beta_j}\Big).$$

Step 2. By (4.13),

$$E\Big(\frac{\partial L_i}{\partial\beta_h}\frac{\partial L_i}{\partial\beta_j}\Big) = \frac{E(Y_i-\mu_i)^2\,x_{ih}x_{ij}}{\mathrm{Var}(Y_i)^2}\Big(\frac{\partial\mu_i}{\partial\eta_i}\Big)^2 = \frac{x_{ih}x_{ij}}{\mathrm{Var}(Y_i)}\Big(\frac{\partial\mu_i}{\partial\eta_i}\Big)^2.$$

Step 3. Summing over the independent observations, $E[-\partial^2 L/\partial\beta_h\partial\beta_j] = \sum_i x_{ih}w_i x_{ij}$, the (h,j) entry of $X^\top W X$. ∎

The estimated asymptotic covariance matrix of $\hat{\boldsymbol{\beta}}$ is therefore

$$\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}}) = \hat{\mathcal{I}}^{-1} = (X^\top \hat{W} X)^{-1}, \tag{4.18}$$

with $\hat{W}$ evaluated at $\hat{\boldsymbol{\beta}}$; W depends on the link. The justification is the following theorem.

Theorem 4.16 (Asymptotic normality of the ML estimator)

Let $\boldsymbol{\beta}_0$ be the true value, interior to the parameter space, and write $\boldsymbol{u}(\boldsymbol{\beta}) = \partial L/\partial \boldsymbol{\beta}$, $H(\boldsymbol{\beta}) = \partial^2 L/\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top$, $\mathcal{I}_N(\boldsymbol{\beta}) = X^\top W(\boldsymbol{\beta}) X$. Assume:
(R1) $N^{-1}\mathcal{I}_N(\boldsymbol{\beta}_0) \to \Sigma$, positive definite;
(R2) Lindeberg condition: for each $\boldsymbol{c} \in \mathbb{R}^p$, the terms $\boldsymbol{c}^\top \partial L_i/\partial\boldsymbol{\beta}$ $(i \le N)$, normalized by $(N\boldsymbol{c}^\top\Sigma\boldsymbol{c})^{1/2}$, satisfy Lindeberg's condition;
(R3) $\sup_{\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le\delta} \|N^{-1}H(\boldsymbol{\beta}) + \Sigma\| \xrightarrow{p} 0$ as $N \to \infty$ and then $\delta \to 0$.
Then with probability tending to one the likelihood equations have a root $\hat{\boldsymbol{\beta}}$ with $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$, and

$$\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0}, \Sigma^{-1}), \qquad \text{i.e.} \qquad \hat{\boldsymbol{\beta}} \approx N_p(\boldsymbol{\beta}_0, (X^\top W X)^{-1}).$$

Proof. [Supplement]

Step 1. (Score CLT.) The summands $\partial L_i/\partial\boldsymbol{\beta}$ are independent with mean $\mathbf{0}$ (first Bartlett identity) and covariance summing to $\mathcal{I}_N(\boldsymbol{\beta}_0)$ (Theorem 4.15). By (R1), (R2), the Lindeberg–Feller CLT and the Cramér–Wold device, $N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0}, \Sigma)$. In particular $N^{-1}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{p} \mathbf{0}$.

Step 2. (Existence of a consistent root.) Let $\lambda > 0$ be the smallest eigenvalue of Σ. Fix small $\delta > 0$ so that, by (R3), with probability $\to 1$, $-N^{-1}H(\boldsymbol{\beta})$ has all eigenvalues $\ge \lambda/2$ on the ball $B_\delta = \{\|\boldsymbol{\beta} - \boldsymbol{\beta}_0\| \le \delta\}$. For $\boldsymbol{\beta}$ on the boundary of B_δ, Taylor's theorem with some $\bar{\boldsymbol{\beta}}$ on the segment gives

$$L(\boldsymbol{\beta}) - L(\boldsymbol{\beta}_0) = \boldsymbol{u}(\boldsymbol{\beta}_0)^\top(\boldsymbol{\beta} - \boldsymbol{\beta}_0) + \tfrac{1}{2}(\boldsymbol{\beta} - \boldsymbol{\beta}_0)^\top H(\bar{\boldsymbol{\beta}})(\boldsymbol{\beta} - \boldsymbol{\beta}_0) \le N\delta\Big(\frac{\|\boldsymbol{u}(\boldsymbol{\beta}_0)\|}{N} - \frac{\lambda\delta}{4}\Big).$$

By Step 1 the right side is negative with probability $\to 1$. Then the maximum of the continuous L over the compact B_δ is attained in the interior, where $\boldsymbol{u} = \mathbf{0}$. Hence a root $\hat{\boldsymbol{\beta}} \in B_\delta$ exists with probability $\to 1$; as δ is arbitrary, $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$.

Step 3. (Expansion.) Applying the mean value theorem to each coordinate of $\boldsymbol{u}$ along the segment from $\boldsymbol{\beta}_0$ to $\hat{\boldsymbol{\beta}}$,

$$\mathbf{0} = \boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \boldsymbol{u}(\boldsymbol{\beta}_0) + \tilde{H}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0),$$

where row j of $\tilde{H}$ is row j of H at an intermediate point. By consistency and (R3), $-N^{-1}\tilde{H} \xrightarrow{p} \Sigma$.

Step 4. Therefore $\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = (-N^{-1}\tilde{H})^{-1}N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0)$, and Slutsky's theorem with Step 1 gives the limit $N_p(\mathbf{0}, \Sigma^{-1}\Sigma\Sigma^{-1}) = N_p(\mathbf{0}, \Sigma^{-1})$. Finally $N^{-1}X^\top\hat{W}X \xrightarrow{p} \Sigma$ by continuity, which justifies (4.18). ∎

For grouped binomial or Poisson data the same argument applies with N fixed and the counts $n_i \to \infty$ (normalize by $n = \sum n_i$ instead of N).

4.4.7 Likelihood equations and covariance matrix for Poisson loglinear model

Corollary 4.17 (Poisson loglinear model)

For $\log\boldsymbol{\mu} = X\boldsymbol{\beta}$ with Poisson responses, the likelihood equations are

$$\sum_i (y_i - \mu_i)x_{ij} = 0, \qquad j = 1, \dots, p, \tag{4.19}$$

and $\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}}) = (X^\top \hat{W} X)^{-1}$ with $\hat{W} = \mathrm{diag}(\hat{\mu}_1, \ldots, \hat{\mu}_N)$.

Proof. For the log link $\mu_i = e^{\eta_i}$, so $\partial \mu_i / \partial \eta_i = \mu_i = \mathrm{Var}(Y_i)$. Then (4.14) becomes (4.19), and $w_i = \mu_i^2/\mu_i = \mu_i$ in (4.17). ■

Equations (4.19) equate the sufficient statistics $\sum_i y_i x_{ij}$ to their expected values (see Theorem 4.27).

4.5 Inference for generalized linear models

Motivation. Once a GLM is fitted, the Wald, score and likelihood-ratio methods apply to any parameter. Likelihood-ratio inference is conveniently organized through the deviance, which plays the role of the residual sum of squares.

Theorem 4.18 (Wald, likelihood-ratio and score tests)

Under the conditions of Theorem 4.16, for $H_0 : \boldsymbol{\beta} = \boldsymbol{\beta}_0$ each of

$$\begin{aligned} W &= (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^\top \hat{\mathcal{I}}\, (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) && \text{(Wald)}, \\ \Lambda &= -2\big[L(\boldsymbol{\beta}_0) - L(\hat{\boldsymbol{\beta}})\big] && \text{(likelihood ratio)}, \\ S &= \boldsymbol{u}(\boldsymbol{\beta}_0)^\top \mathcal{I}(\boldsymbol{\beta}_0)^{-1} \boldsymbol{u}(\boldsymbol{\beta}_0) && \text{(score)} \end{aligned}$$

converges in distribution to χ^2_p, and the three differ by $o_p(1)$. For nested models $M_0 \subset M_1$, where M_0 imposes q linear restrictions on $\boldsymbol{\beta}$, $\Lambda = -2[L(\hat{\boldsymbol{\beta}}_0) - L(\hat{\boldsymbol{\beta}}_1)] \xrightarrow{d} \chi^2_q$.

Proof. [Supplement] Let $\boldsymbol{Z}_N = N^{-1/2} \boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} \boldsymbol{Z} \sim N_p(\mathbf{0}, \Sigma)$ and, from Steps 3–4 of Theorem 4.16, $\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$.

Step 1. (Score.) $S = \boldsymbol{Z}_N^\top (N^{-1}\mathcal{I}_N)^{-1} \boldsymbol{Z}_N \xrightarrow{d} \boldsymbol{Z}^\top \Sigma^{-1} \boldsymbol{Z} \sim \chi^2_p$, since $\Sigma^{-1/2} \boldsymbol{Z} \sim N_p(\mathbf{0}, I)$.

Step 2. (Wald.) $W = [\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)]^\top (N^{-1}\hat{\mathcal{I}}) [\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)] = \boldsymbol{Z}_N^\top \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$.

Step 3. (LR.) Expand L about $\hat{\boldsymbol{\beta}}$, where $\boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$:

$$L(\boldsymbol{\beta}_0) = L(\hat{\boldsymbol{\beta}}) + \tfrac{1}{2} (\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})^\top H(\bar{\boldsymbol{\beta}}) (\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}}),$$

so $\Lambda = [\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)]^\top (-N^{-1} H(\bar{\boldsymbol{\beta}})) [\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)] = \boldsymbol{Z}_N^\top \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$ by (R3).

Step 4. (Nested models.) Write M_0 as $\boldsymbol{\beta} = \boldsymbol{\beta}_0 + A\boldsymbol{\gamma}$ with A a $p \times (p-q)$ matrix of full rank. Applying Step 3 to M_1 and to the reparametrized model M_0 (whose score is $A^\top \boldsymbol{u}$ and information $A^\top \mathcal{I} A$),

$$\Lambda = \boldsymbol{Z}_N^\top \big[\Sigma^{-1} - A(A^\top \Sigma A)^{-1} A^\top\big] \boldsymbol{Z}_N + o_p(1).$$

With $\boldsymbol{Z} = \Sigma^{1/2} \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N_p(\mathbf{0}, I)$, the limit is $\boldsymbol{\varepsilon}^\top (I - P) \boldsymbol{\varepsilon}$, where $P = \Sigma^{1/2} A (A^\top \Sigma A)^{-1} A^\top \Sigma^{1/2}$ is a symmetric idempotent matrix of rank $p - q$. Thus $I - P$ is a projection of rank q, and $\boldsymbol{\varepsilon}^\top (I-P) \boldsymbol{\varepsilon} \sim \chi^2_q$. ■

Confidence regions follow by inverting these tests; e.g. the Wald interval $\hat{\beta}_j \pm z_{\alpha/2} \widehat{\mathrm{se}}(\hat{\beta}_j)$ and the profile-likelihood interval $\{\beta_j : \Lambda(\beta_j) \le \chi^2_1(\alpha)\}$.

4.5.1 Deviance and goodness of fit

Let $\tilde{\theta}_i$ be the natural parameter of the saturated model ($\tilde{\mu}_i = y_i$) and $\hat{\theta}_i$ that of the fitted model. By (4.12) with $a(\phi) = \phi/\omega_i$,

$$-2\big[L(\hat{\boldsymbol{\mu}}; \mathbf{y}) - L(\mathbf{y}; \mathbf{y})\big] = 2\sum_i \frac{\omega_i\big[y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)\big]}{\phi}.$$

Definition (Scaled deviance and deviance)

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2\sum_i \omega_i\big[y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)\big] \tag{4.20}$$

is the *deviance*, and $D(\mathbf{y}; \hat{\boldsymbol{\mu}})/\phi$ the *scaled deviance*. The larger it is, the poorer the fit. For Poisson and binomial models $\phi = 1$, and the two coincide.

Theorem 4.19 (Asymptotic distribution of the deviance)

For a binomial or Poisson GLM with a fixed number N of observations whose counts grow ($n_i \to \infty$, or $\mu_i \to \infty$), if the model with p parameters holds then $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) \xrightarrow{d} \chi^2_{N-p}$.

Proof. [Supplement] The saturated model is a GLM with N parameters (one per observation, θ_i free), and the working model is the nested submodel obtained by $N-p$ linear restrictions $\boldsymbol{\theta} \in \mathrm{col}(X)$ (canonical scale; for other links apply a smooth reparametrization). The deviance is $-2[L(\hat{\boldsymbol{\mu}}) - L(\tilde{\boldsymbol{\mu}})]$, the likelihood-ratio statistic for these nested models; Theorem 4.18 (with the growing counts playing the role of sample size) gives the χ^2_{N-p} limit. ■

Normal deviance

For the normal linear model ($b(\theta) = \theta^2/2$, $\omega_i = 1$), $D = \sum_i (y_i - \hat{\mu}_i)^2$, the residual sum of squares, and D/σ^2 is the scaled deviance.

4.5.2 Deviance for Poisson models

Proposition 4.20 (Poisson deviance)

For a Poisson GLM,

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2\sum_i \big[y_i \log(y_i/\hat{\mu}_i) - y_i + \hat{\mu}_i\big]. \tag{4.21}$$

If the model has log link and an intercept, $\sum_i y_i = \sum_i \hat{\mu}_i$ and $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2\sum_i y_i \log(y_i/\hat{\mu}_i)$.

Proof. Here $\hat{\theta}_i = \log \hat{\mu}_i$, $b(\hat{\theta}_i) = \hat{\mu}_i$, $\tilde{\theta}_i = \log y_i$, $b(\tilde{\theta}_i) = y_i$, $\omega_i = 1$; (4.20) gives (4.21). With an intercept ($x_{i1} \equiv 1$), the first equation of (4.19) is $\sum_i (y_i - \hat{\mu}_i) = 0$, so the linear terms cancel. (Terms with $y_i = 0$ are 0, using $0 \log 0 = 0$.) ■

For a two-way table and the independence fit $\hat{\mu}_{ij}$, this is G^2 of Chapter 3. For a fixed number of cells and large means, the deviance is approximately chi-squared (Section 14.3).

4.5.3 Deviance for binomial models: grouped and ungrouped data

Proposition 4.21 (Binomial deviance)

For binomial proportions y_i based on n_i trials,

$$D(\mathbf{y};\hat{\boldsymbol{\mu}}) = 2\sum_i n_i y_i \log\frac{n_i y_i}{n_i\hat{\pi}_i} + 2\sum_i (n_i - n_i y_i)\log\frac{n_i - n_i y_i}{n_i - n_i\hat{\pi}_i}, \tag{4.22}$$

that is, $D = 2\sum \text{observed} \times \log(\text{observed}/\text{fitted})$ over the $2N$ cells of successes and failures.

Proof. Here $\hat{\theta}_i = \operatorname{logit}\hat{\pi}_i$, $b(\hat{\theta}_i) = -\log(1-\hat{\pi}_i)$, $\tilde{\theta}_i = \operatorname{logit} y_i$, $b(\tilde{\theta}_i) = -\log(1-y_i)$, $\omega_i = n_i$. By (4.20),

$$\begin{aligned} D &= 2\sum_i n_i\Big[y_i\log\frac{y_i}{1-y_i} - y_i\log\frac{\hat{\pi}_i}{1-\hat{\pi}_i} + \log(1-y_i) - \log(1-\hat{\pi}_i)\Big] \\ &= 2\sum_i n_i\Big[y_i\log\frac{y_i}{\hat{\pi}_i} + (1-y_i)\log\frac{1-y_i}{1-\hat{\pi}_i}\Big], \end{aligned}$$

which is (4.22) after multiplying numerator and denominator by n_i. ■

Grouped versus ungrouped data

Binomial data may be entered as success/failure counts at each predictor setting (*grouped*) or as individual 0/1 observations (*ungrouped*). The ML estimates agree, but the saturated model and hence the deviance differ. The chi-squared approximation for the deviance holds for grouped data with the number of settings fixed and $n_i \to \infty$, not for ungrouped data, where the number of saturated-model parameters grows with n.

4.5.4 Likelihood-ratio model comparison using the deviance

Proposition 4.22 (Deviance difference)

Let $M_0 \subset M_1$ be nested Poisson or binomial GLMs with fitted values $\hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\mu}}_1$. Then $D(\mathbf{y};\hat{\boldsymbol{\mu}}_1) \le D(\mathbf{y};\hat{\boldsymbol{\mu}}_0)$ and the likelihood-ratio statistic for $H_0 : M_0$ given M_1 is

$$-2\big[L(\hat{\boldsymbol{\mu}}_0;\mathbf{y}) - L(\hat{\boldsymbol{\mu}}_1;\mathbf{y})\big] = D(\mathbf{y};\hat{\boldsymbol{\mu}}_0) - D(\mathbf{y};\hat{\boldsymbol{\mu}}_1),$$

asymptotically χ^2 with df equal to the difference in numbers of parameters. With log link and intercept (Poisson) or for binomial data, it equals $2\sum \text{observed} \times \log(\text{fitted}_1/\text{fitted}_0)$.

Proof. Maximizing over the smaller parameter space cannot give a larger maximum, so $L(\hat{\boldsymbol{\mu}}_0) \le L(\hat{\boldsymbol{\mu}}_1)$. The identity follows by adding and subtracting $L(\mathbf{y};\mathbf{y})$, which is the same for both models. The distribution is Theorem 4.18. For the last form, subtract (4.22) (or (4.21) without linear terms) for the two models: the log observed terms cancel. ■

Since the saturated part cancels, the comparison does not depend on whether binomial data are grouped or ungrouped, and it can be valid even when each deviance alone is not approximately chi-squared.

4.5.5 Residuals for GLMs

Definition (Deviance and Pearson residuals)

Write $D(\mathbf{y};\hat{\boldsymbol{\mu}}) = \sum_i d_i$ with $d_i = 2\omega_i[y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)]$. The *deviance residual* is $\sqrt{d_i}\,\operatorname{sgn}(y_i - \hat{\mu}_i)$, and the *Pearson residual* is

$$e_i = \frac{y_i - \hat{\mu}_i}{[\widehat{\operatorname{Var}}(Y_i)]^{1/2}} \qquad (\text{Poisson: } e_i = (y_i - \hat{\mu}_i)/\sqrt{\hat{\mu}_i}).$$

For the independence model in a two-way table, $\sum e_{ij}^2 = X^2$ and $\sum d_{ij} = G^2$. These residuals are less variable than $N(0,1)$ because they compare y_i with $\hat{\mu}_i$ rather than μ_i. Standardizing uses the *hat matrix*

$$\mathrm{HatM} = W^{1/2}X(X^\top WX)^{-1}X^\top W^{1/2}. \tag{4.23}$$

Proposition 4.23 (Leverages and residual covariance)

(a) HatM is symmetric and idempotent with trace p; its diagonal elements (*leverages*) satisfy $0 \le h_i \le 1$ and $\sum_i h_i = p$.
(b) To first order, with $V = \operatorname{Cov}(\boldsymbol{Y}) = \operatorname{diag}\{\operatorname{Var}(Y_i)\}$,

$$\operatorname{Cov}(\boldsymbol{Y} - \hat{\boldsymbol{\mu}}) \approx V^{1/2}(I - \mathrm{HatM})V^{1/2}, \qquad \operatorname{Var}(Y_i - \hat{\mu}_i) \approx \operatorname{Var}(Y_i)(1 - h_i).$$

Hence the *standardized Pearson residual* is

$$r_i = \frac{y_i - \hat{\mu}_i}{[\widehat{\operatorname{Var}}(Y_i)(1-\hat{h}_i)]^{1/2}} = \frac{e_i}{\sqrt{1-\hat{h}_i}} \qquad (\text{Poisson: } \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i(1-\hat{h}_i)}}). \tag{4.24}$$

Proof. [Supplement] (a) Let $A = W^{1/2}X$. Then $\mathrm{HatM} = A(A^\top A)^{-1}A^\top$ is symmetric with $\mathrm{HatM}^2 = \mathrm{HatM}$ and $\operatorname{tr}\mathrm{HatM} = \operatorname{tr}[(A^\top A)^{-1}A^\top A] = \operatorname{tr} I_p = p$. By symmetry and idempotence $h_i = \sum_j h_{ij}^2 \ge h_i^2$, so $0 \le h_i \le 1$.

(b)

Step 1. Let $G = \operatorname{diag}(\partial\eta_i/\partial\mu_i)$. Since $\partial\mu_i/\partial\eta_i = 1/G_{ii}$, (4.13) gives $\boldsymbol{u} = X^\top WG(\boldsymbol{y} - \boldsymbol{\mu})$.

Step 2. From Theorem 4.16, $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \approx (X^\top WX)^{-1}X^\top WG(\boldsymbol{y} - \boldsymbol{\mu})$, and $\hat{\boldsymbol{\mu}} - \boldsymbol{\mu} \approx G^{-1}X(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$. Hence

$$\boldsymbol{y} - \hat{\boldsymbol{\mu}} \approx [I - G^{-1}X(X^\top WX)^{-1}X^\top WG](\boldsymbol{y} - \boldsymbol{\mu}).$$

Step 3. Because $w_i = 1/(G_{ii}^2 \operatorname{Var}(Y_i))$, we have $V = G^{-2}W^{-1}$, i.e. $V^{1/2} = G^{-1}W^{-1/2} =: M$. Then $G^{-1}X(X^\top WX)^{-1}X^\top WG = M\,\mathrm{HatM}\,M^{-1}$, so $\boldsymbol{y} - \hat{\boldsymbol{\mu}} \approx M(I - \mathrm{HatM})M^{-1}(\boldsymbol{y} - \boldsymbol{\mu})$.

Step 4. With $\operatorname{Cov}(\boldsymbol{y} - \boldsymbol{\mu}) = V = M^2$ and HatM symmetric idempotent,

$$\operatorname{Cov}(\boldsymbol{y} - \hat{\boldsymbol{\mu}}) \approx M(I - \mathrm{HatM})M^{-1}M^2M^{-1}(I - \mathrm{HatM})M = M(I - \mathrm{HatM})M.$$

Its ith diagonal entry is $\operatorname{Var}(Y_i)(1 - h_i)$. ∎

As in linear models, larger leverage means greater potential influence. Unlike ordinary regression, the hat values depend on the fit through W, so extreme predictor values need not have high leverage. Applied to a linearized $g(\mathbf{y})$, HatM yields (after rescaling) the fitted linear predictor $\hat{\boldsymbol{\eta}}$.

4.6 Fitting generalized linear models

Motivation. The likelihood equations (4.14) are nonlinear in $\boldsymbol{\beta}$. Newton–Raphson and Fisher scoring solve them iteratively; Fisher scoring turns out to be a sequence of weighted least squares fits, which is how all GLM software works.

4.6.1 Newton–Raphson method

Let $\boldsymbol{u} = \partial L/\partial \boldsymbol{\beta}$ and $H = \partial^2 L/\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top$ (the Hessian), and let $\boldsymbol{u}^{(t)}, H^{(t)}$ be their values at the guess $\boldsymbol{\beta}^{(t)}$. Near $\boldsymbol{\beta}^{(t)}$,

$$L(\boldsymbol{\beta}) \approx L(\boldsymbol{\beta}^{(t)}) + \boldsymbol{u}^{(t)\top}(\boldsymbol{\beta} - \boldsymbol{\beta}^{(t)}) + \tfrac{1}{2}(\boldsymbol{\beta} - \boldsymbol{\beta}^{(t)})^\top H^{(t)}(\boldsymbol{\beta} - \boldsymbol{\beta}^{(t)}).$$

Setting the gradient of this quadratic, $\boldsymbol{u}^{(t)} + H^{(t)}(\boldsymbol{\beta} - \boldsymbol{\beta}^{(t)})$, to $\mathbf{0}$ gives the *Newton–Raphson* step

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - (H^{(t)})^{-1}\boldsymbol{u}^{(t)}, \tag{4.25}$$

assuming $H^{(t)}$ nonsingular (in practice one solves the linear system rather than inverting). Iterate until L changes negligibly. With several local maxima a good starting value is crucial. Figure 2 shows one cycle.

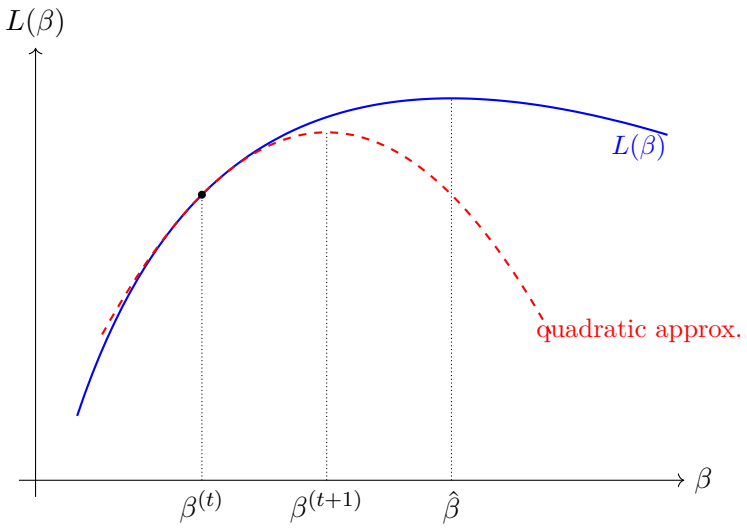


Figure 2: One Newton–Raphson cycle: the next guess maximizes the second-order Taylor approximation at $\beta^{(t)}$.

Example 4.24 (Newton–Raphson for a binomial parameter)

For one $\mathrm{bin}(n, \pi)$ observation y, $L(\pi) = y\log\pi + (n-y)\log(1-\pi)$, so

$$u = \frac{y - n\pi}{\pi(1-\pi)}, \qquad H = -\Big[\frac{y}{\pi^2} + \frac{n-y}{(1-\pi)^2}\Big],$$

$$\pi^{(t+1)} = \pi^{(t)} + \Big[\frac{y}{\pi^{(t)2}} + \frac{n-y}{(1-\pi^{(t)})^2}\Big]^{-1} \frac{y - n\pi^{(t)}}{\pi^{(t)}(1-\pi^{(t)})}.$$

The step moves $\pi^{(t)}$ up if $y/n > \pi^{(t)}$ and down otherwise; at $\pi^{(t)} = y/n$ no adjustment occurs. Starting from $\pi^{(0)} = \frac{1}{2}$ one gets $\pi^{(1)} = y/n$ at once; from other starting values four or five iterations usually suffice.

Theorem 4.25 (Quadratic convergence of Newton–Raphson)

Suppose $\boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$, H is Lipschitz on a ball B around $\hat{\boldsymbol{\beta}}$ with constant γ ($\|H(\boldsymbol{a}) - H(\boldsymbol{b})\| \le \gamma\|\boldsymbol{a} - \boldsymbol{b}\|$), and $\|H(\boldsymbol{\beta})^{-1}\| \le M$ on B. If $\boldsymbol{\beta}^{(t)} \in B$, then

$$\|\boldsymbol{\beta}^{(t+1)} - \hat{\boldsymbol{\beta}}\| \le \frac{M\gamma}{2}\,\|\boldsymbol{\beta}^{(t)} - \hat{\boldsymbol{\beta}}\|^2.$$

In particular, if $\boldsymbol{\beta}^{(0)} \in B$ and $\|\boldsymbol{\beta}^{(0)} - \hat{\boldsymbol{\beta}}\| < 2/(M\gamma)$, the iterates converge to $\hat{\boldsymbol{\beta}}$ and the number of correct digits roughly doubles each step.

Proof. [Supplement]

Step 1. Let $\boldsymbol{e}_t = \boldsymbol{\beta}^{(t)} - \hat{\boldsymbol{\beta}}$. Since $\boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$,

$$\boldsymbol{u}^{(t)} = \boldsymbol{u}^{(t)} - \boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \int_0^1 H(\hat{\boldsymbol{\beta}} + s\boldsymbol{e}_t)\,ds\;\boldsymbol{e}_t.$$

Step 2. By (4.25),

$$\boldsymbol{e}_{t+1} = \boldsymbol{e}_t - (H^{(t)})^{-1}\boldsymbol{u}^{(t)} = (H^{(t)})^{-1}\int_0^1 [H(\hat{\boldsymbol{\beta}} + \boldsymbol{e}_t) - H(\hat{\boldsymbol{\beta}} + s\boldsymbol{e}_t)]ds\;\boldsymbol{e}_t.$$

Step 3. The integrand has norm at most $\gamma(1-s)\|\boldsymbol{e}_t\|$, and $\int_0^1 (1-s)\,ds = \frac{1}{2}$. Hence $\|\boldsymbol{e}_{t+1}\| \le M \cdot \frac{\gamma}{2}\|\boldsymbol{e}_t\|^2$.

Step 4. If $\rho = \frac{M\gamma}{2}\|\boldsymbol{e}_0\| < 1$, then by induction $\|\boldsymbol{e}_{t+1}\| \le \rho\|\boldsymbol{e}_t\|$, the iterates stay in B, and $\|\boldsymbol{e}_t\| \to 0$; the quadratic bound then shows the error is roughly squared at each step. ∎

4.6.2 Fisher scoring method

Fisher scoring replaces the Hessian (minus the *observed information*) by its expectation, minus the *expected information* $\mathcal{I}$. With $\mathcal{I}^{(t)}$ evaluated at $\boldsymbol{\beta}^{(t)}$,

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + (\mathcal{I}^{(t)})^{-1}\boldsymbol{u}^{(t)} \quad\Longleftrightarrow\quad \mathcal{I}^{(t)}\boldsymbol{\beta}^{(t+1)} = \mathcal{I}^{(t)}\boldsymbol{\beta}^{(t)} + \boldsymbol{u}^{(t)}. \tag{4.26}$$

For the binomial parameter, $\mathcal{I} = n/[\pi(1-\pi)]$ and

$$\pi^{(t+1)} = \pi^{(t)} + \frac{\pi^{(t)}(1-\pi^{(t)})}{n} \cdot \frac{y - n\pi^{(t)}}{\pi^{(t)}(1-\pi^{(t)})} = \frac{y}{n},$$

the answer after one step from any start. For a GLM, $\mathcal{I}^{(t)} = X^\top W^{(t)} X$ by Theorem 4.15, and $\boldsymbol{u}^{(t)}$ has elements (4.14):

$$u_j = \sum_{i=1}^{N} \frac{(y_i - \mu_i)x_{ij}}{\mathrm{Var}(Y_i)}\frac{\partial \mu_i}{\partial \eta_i}. \tag{4.27}$$

At convergence $(\mathcal{I}^{(t)})^{-1}$ is the estimated covariance (4.18), a free by-product.

Observed or expected information?

For canonical links the two coincide (Theorem 4.27). Otherwise Fisher scoring has the advantages that $\mathcal{I}$ is automatically nonnegative definite, it yields the covariance matrix, and it is weighted least squares (next subsection); but it need not converge quadratically, and for complex models the observed information may be easier to compute. Efron and Hinkley (1978) argued that observed-information variances better approximate the relevant conditional vari-

ance and agree more closely with Bayesian analyses.

4.6.3 ML as iterative reweighted least squares

For the linear model $\boldsymbol{z} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathrm{Cov}(\boldsymbol{\varepsilon}) = V$, the weighted least squares (WLS) estimator is $(X^\top V^{-1} X)^{-1} X^\top V^{-1} \boldsymbol{z}$.

Theorem 4.26 (Fisher scoring is IRLS)

For a GLM, the Fisher scoring step (4.26) is

$$\boldsymbol{\beta}^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} \boldsymbol{z}^{(t)},$$

the WLS fit of the *working response*

$$z_i^{(t)} = \eta_i^{(t)} + (y_i - \mu_i^{(t)}) \frac{\partial \eta_i^{(t)}}{\partial \mu_i^{(t)}} \tag{4.28}$$

on X with weight matrix (inverse covariance) $W^{(t)}$.

Proof.

Step 1. By Theorem 4.15, $\mathcal{I}^{(t)} \boldsymbol{\beta}^{(t)} = X^\top W^{(t)} X \boldsymbol{\beta}^{(t)} = X^\top W^{(t)} \boldsymbol{\eta}^{(t)}$.

Step 2. Since $w_i = (\partial \mu_i / \partial \eta_i)^2 / \operatorname{Var}(Y_i)$, each term of (4.27) is

$$\frac{(y_i - \mu_i) x_{ij}}{\operatorname{Var}(Y_i)} \frac{\partial \mu_i}{\partial \eta_i} = x_{ij}\, w_i\, (y_i - \mu_i) \frac{\partial \eta_i}{\partial \mu_i},$$

so $\boldsymbol{u}^{(t)} = X^\top W^{(t)} \boldsymbol{d}^{(t)}$ with $d_i^{(t)} = (y_i - \mu_i^{(t)})\, \partial \eta_i^{(t)} / \partial \mu_i^{(t)}$.

Step 3. Adding, the right side of (4.26) is $X^\top W^{(t)} (\boldsymbol{\eta}^{(t)} + \boldsymbol{d}^{(t)}) = X^\top W^{(t)} \boldsymbol{z}^{(t)}$. Thus $X^\top W^{(t)} X \boldsymbol{\beta}^{(t+1)} = X^\top W^{(t)} \boldsymbol{z}^{(t)}$, the WLS normal equations. ∎

The working response is the linearization of the link at the data,

$$g(y_i) \approx g(\mu_i) + (y_i - \mu_i) g'(\mu_i) = \eta_i + (y_i - \mu_i) \frac{\partial \eta_i}{\partial \mu_i} = z_i,$$

and W^{-1} is approximately its covariance matrix. Each cycle regresses $\boldsymbol{z}^{(t)}$ on X with weights $W^{(t)}$, giving $\boldsymbol{\beta}^{(t+1)}$, then $\boldsymbol{\eta}^{(t+1)} = X\boldsymbol{\beta}^{(t+1)}$ and new $\boldsymbol{z}^{(t+1)}$, $W^{(t+1)}$: *iteratively reweighted least squares* (IRLS). A simple start uses $\hat{\boldsymbol{\mu}}^{(0)} = \mathbf{y}$, adjusting values where $g(y_i)$ is infinite (e.g. $y_i = 0$ with log link: use $y_i = \frac{1}{2}$ in the first cycle only).

4.6.4 Simplifications for canonical links

Theorem 4.27 (Canonical link properties)

Consider a GLM with the canonical link, $\theta_i = \eta_i = \sum_j \beta_j x_{ij}$, and a common $a(\phi)$ for all i (e.g. Poisson; binomial with all $n_i = 1$). Then:
(a) $\sum_i y_i x_{ij}$, $j = 1, \ldots, p$, are sufficient statistics for $\boldsymbol{\beta}$;
(b) $\partial L_i / \partial \beta_j = (y_i - \mu_i) x_{ij} / a(\phi)$, so the likelihood equations are

$$\sum_i x_{ij} y_i = \sum_i x_{ij} \mu_i, \qquad j = 1, \ldots, p, \qquad \text{i.e.} \quad X^\top \mathbf{y} = X^\top \boldsymbol{\mu}; \tag{4.29}$$

(c) $\partial^2 L/\partial\beta_h\partial\beta_j = -\sum_i x_{ih}x_{ij}b''(\theta_i)/a(\phi)$ does not involve $\mathbf{y}$; hence observed and expected information coincide, $H = -\mathcal{I}$, and Newton–Raphson equals Fisher scoring;
(d) L is concave in $\boldsymbol{\beta}$, strictly if X has full column rank; so a solution of (4.29), if one exists, is the unique ML estimate.
With weights $a(\phi) = \phi/\omega_i$ the same holds with y_i, μ_i replaced by $\omega_i y_i, \omega_i \mu_i$.

Proof.

Step 1. By (4.12) the part of L involving both data and parameters is

$$\frac{1}{a(\phi)}\sum_i y_i\theta_i = \frac{1}{a(\phi)}\sum_i y_i \sum_j \beta_j x_{ij} = \frac{1}{a(\phi)}\sum_j \beta_j\Big(\sum_i y_i x_{ij}\Big),$$

and the rest depends on $\boldsymbol{\beta}$ only or on $\mathbf{y}$ only; the factorization theorem gives (a).

Step 2. For the canonical link $\partial\mu_i/\partial\eta_i = \partial\mu_i/\partial\theta_i = b''(\theta_i) = \mathrm{Var}(Y_i)/a(\phi)$; inserting into (4.13) gives $\partial L_i/\partial\beta_j = (y_i - \mu_i)x_{ij}/a(\phi)$, and (b) follows.

Step 3. Differentiating again, $\partial\mu_i/\partial\beta_h = b''(\theta_i)x_{ih}$, so $\partial^2 L_i/\partial\beta_h\partial\beta_j = -x_{ij}x_{ih}b''(\theta_i)/a(\phi)$, free of y_i; its expectation is itself, giving (c).

Step 4. For any $\boldsymbol{c} \in \mathbb{R}^p$, $\boldsymbol{c}^\top H\boldsymbol{c} = -\sum_i b''(\theta_i)(\boldsymbol{x}_i^\top \boldsymbol{c})^2/a(\phi) \le 0$ since $b'' > 0$; equality forces $X\boldsymbol{c} = \mathbf{0}$, impossible for $\boldsymbol{c} \neq \mathbf{0}$ under full rank. So L is (strictly) concave; a stationary point of a strictly concave function is its unique global maximum, giving (d). ■

Equations (4.29) equate sufficient statistics to their expected values (Nelder and Wedderburn 1972): for the normal linear model they are the normal equations; for Poisson loglinear models they are (4.19); for binary logistic regression they are (4.16) with $n_i = 1$.